COURSE TITLE : FLUID MECHANICS LAB
COURSE CODE : 4078
COURSE CATEGORY : B
PERIODS/ WEEK : 6
PERIODS/ SEMESTER : 84
CREDIT : 3

TIME SCHEDULE

MODULE	TOPIC	PERIOD
1	Measurement of pipe Friction	14
2	Flow Measurement	28
3	Efficiency Determination	28
4	Velocity Distribution	14
TOTAL		**84**

COURSE OUTCOME :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	Determine the coefficient of discharge of venturimeter.
	2	Determine the coefficient of discharge of orificemeter.
	3	Calibrate a rotameter and plot a calibration curve.
2	4	To perform experiment on Bernoulli's theorem and prove that the summation of pressure head, kinetic head and potential head is constant.
	5	Perform Reynolds experiment.
	6	Plot the characteristics curves of centrifugal pump.
3	7	Determine the viscosity of different liquids.

SPECIFIC OUTCOMES

MODULE – I

1.1.0 Know pipe Friction Apparatus

1.1.1. Determine pipe friction in straight pipe
1.1.2. Determine pipe friction in straight circular pipe with fluid air using Fanning'sequation
1.1.3. Verification of Fanning's equation with pipe fittings

MODULE – II

2.1.0 Understand Flow Measurement

2.1.1. Determine Venturi coefficient
2.1.2. Determine Orifice coefficient

2.1.3. Calibrate the given Rota meter
2.1.4. Determine flow through Weirs and Notches (Rectangular & V-Notch)

MODULE- III

3.1.0 Efficiency Determination

3.1.1. Determine efficiency of centrifugal pump
3.1.2. Determine efficiency of gear pump
3.1.3. Determine efficiency of blower
3.1.4. Determine Volumetric efficiency of compressor

MODULE – IV

4.1.0 Know Velocity Distribution

4.1.1. Determine the velocity profile of airflow in straight pipe
4.1.2. Determine point velocity using Pitot tube
4.1.3. Conduct Reynolds experiment
4.1.4. Determine orifice coefficient using open orifice
4.1.5 Determine the drop in pressure through a packed column
4.1.6 Determine the drop in pressure through a Fluidization column
4.1.7 Verify Stroke's law

REFERENCE

Walter.L.Badger&Julius.T.Banchero	-	Introduction to chemical Engg
Warren.L.McCabe&Julian.C.Smith	-	Unit Operations
K.A.Gavhane	-	Unit Operations – I